

FIFA World Cup Match Predictor

A Player-Aware Machine Learning Framework for
International Match Outcome Prediction

Project Report
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11 - Hussain Cochinwala 15 - Aditya Dhuri 23 - Abhishek Joshi

Prepared following the methodology of Al-Bustami and Ghazal, "From Players to Champions: A Generalizable Machine Learning Approach for Match Outcome Prediction with Insights from the FIFA World Cup," 2025 IEEE International Conference on Electro Information Technology (eIT), and scoped to the data window defined in the project proposal.

Headline result: an 82.2% accuracy ensemble against a 75.4% baseline on 281 World Cup qualifying matches withheld from every development decision (McNemar $p = 0.0134$).

Abstract

National squads are rebuilt before every World Cup, so a team's past results are a weak signal for who is actually on the pitch. This project fuses player-level squad ratings, aggregated into a profile for the specific tournament edition being predicted, with historical team form built from the full international match record, following the methodology of Al-Bustami and Ghazal [1]. A five-model majority-vote ensemble is benchmarked against the paper's Weighted Win Ratio baseline, with a Poisson plus Dixon-Coles [2] scoreline model as a secondary output. Because roughly fifteen configurations were compared against the same development splits over the course of the project, a full season (2025, 281 matches) was withheld from every development decision and scored exactly once: the ensemble reaches 82.2% accuracy against the baseline's 75.4%, a statistically significant margin (McNemar $p = 0.0134$). On development data across five seeds it scores $79.0\% \pm 3.3\%$ against 76.0%, and an independent backtest against the real 2026 World Cup knockout bracket reaches 83.3% against 66.7%. The system is delivered as a tested, end-to-end pipeline with a REST API and web front end.

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1. Introduction

1.1 Background and Motivation

Predicting the outcome of international football matches is a long-standing problem in sports analytics, and it has taken on renewed interest as machine learning methods have matured. Most published approaches rely on team-level signals only: FIFA rankings, historical results, and aggregate market value. These signals work reasonably well in domestic league settings, where squads are stable across a season, but they are a poor fit for a World Cup. National squads are assembled fresh before each tournament from players spread across different clubs and leagues, so a team's reputation two years ago says little about the players who will actually take the field. A model that only sees team names and past results cannot represent a squad that has visibly strengthened or weakened between cycles.

This project follows the approach proposed by Al-Bustami and Ghazal [1], which closes that gap by fusing two categories of information that are usually modelled separately: player-level attributes, aggregated into a squad profile for the specific tournament edition being predicted, and team-level historical form, built from the full international match record. The implementation described in this report follows that paper's structure while adapting it to the data actually available for this project's scope, and extends it with a from-scratch history-feature engine, a formal held-out evaluation protocol, and a working web application.

1.2 Problem Statement

Given two national teams and a FIFA game edition, predict which team wins a match played between them, using only information that would have been available before kickoff. The task is framed as it is in the reference paper [1]: a binary classification over Team A win / Team B win, with draws excluded from the classifier's training target (a secondary model, described in Section 6.4, restores draws and predicts a full scoreline instead). The central methodological question this project had to answer is not just "how accurate can the model be," but "how would we know if the answer were wrong", since the same data was used repeatedly across many rounds of feature and model selection. Section 7 describes how that question shaped the evaluation protocol.

1.3 Objectives

- Build year-specific team profiles from player attribute data, so that squad turnover between tournament cycles is reflected directly in the model's inputs rather than smoothed away by a team-name identifier.
- Engineer match-level features that combine those squad profiles with historical form, Elo rating, and head-to-head record, computed strictly as of each match date to avoid look-ahead leakage.
- Train and evaluate a five-model ensemble against a domain-specific baseline (the paper's Weighted Win Ratio model), under a train/test protocol that is resistant to the model-selection bias that comes from repeatedly tuning against the same test data.
- Extend the classification task with a Poisson-based scoreline model, so the system can express expected goals and a full outcome distribution rather than a single win/lose label.
- Deliver a working, end-to-end demonstration: a cached, config-driven training pipeline, a REST API, and a web interface for live predictions and result exploration.

1.4 Scope and Report Structure

The project covers data collection through to a served web application; it does not cover productionising the system beyond a local demonstration. Section 2 places the project relative to prior work. Section 3 gives the system's architecture as actually built. Sections 4 through 6 describe the data, feature engineering, and models in detail. Section 7 explains the evaluation protocol and why a formal holdout was introduced partway through the project. Section 8 reports results, including the headline held-out figure, the development-time seed variance, and two independent validity checks. Sections 9 and 10 cover the served application and test coverage. Sections 11 through 13 close with limitations, future work, and a conclusion.

2. Related Work

Team-level approaches dominate the published literature: logistic regression and decision trees over historical results and betting-derived features, random forests over aggregated league statistics, and probabilistic team-strength models such as the ones used for World Cup forecasting by national statistics groups, as surveyed in the reference paper's own review of prior work [1]. These methods are effective within a single competition where rosters are stable, but they treat a national team as a single evolving entity rather than a specific set of players, which is exactly the assumption a World Cup breaks.

A smaller body of work incorporates player-specific statistics, such as goals, assists, and passing accuracy, and shows that individual attributes carry information aggregate team statistics do not. Graph-based methods that model player interaction explicitly have shown promise in other sports (notably American football) but remain largely unexplored for international football, in part because the roster and formation data such methods need is harder to source consistently across nations and years [1]. This project's reference paper [1] positions itself at the intersection of these two lines of work: player-level features, aggregated to the team level for tractability, combined with the historical signal that team-level models already use well. The implementation in this report follows that same position, and where it diverges from the paper (a from-scratch history-feature engine, a different leakage-safe evaluation protocol, an explicit scoreline model) those differences and the reasoning behind them are called out in the relevant sections.

3. System Architecture

The system is organised as a pipeline: raw data is transformed into a trained, cached model bundle, which is then served through a REST API and consumed by a web front end. Figure 1 shows the full path, naming the actual module or file responsible for each stage, verified directly against the codebase rather than described from memory.

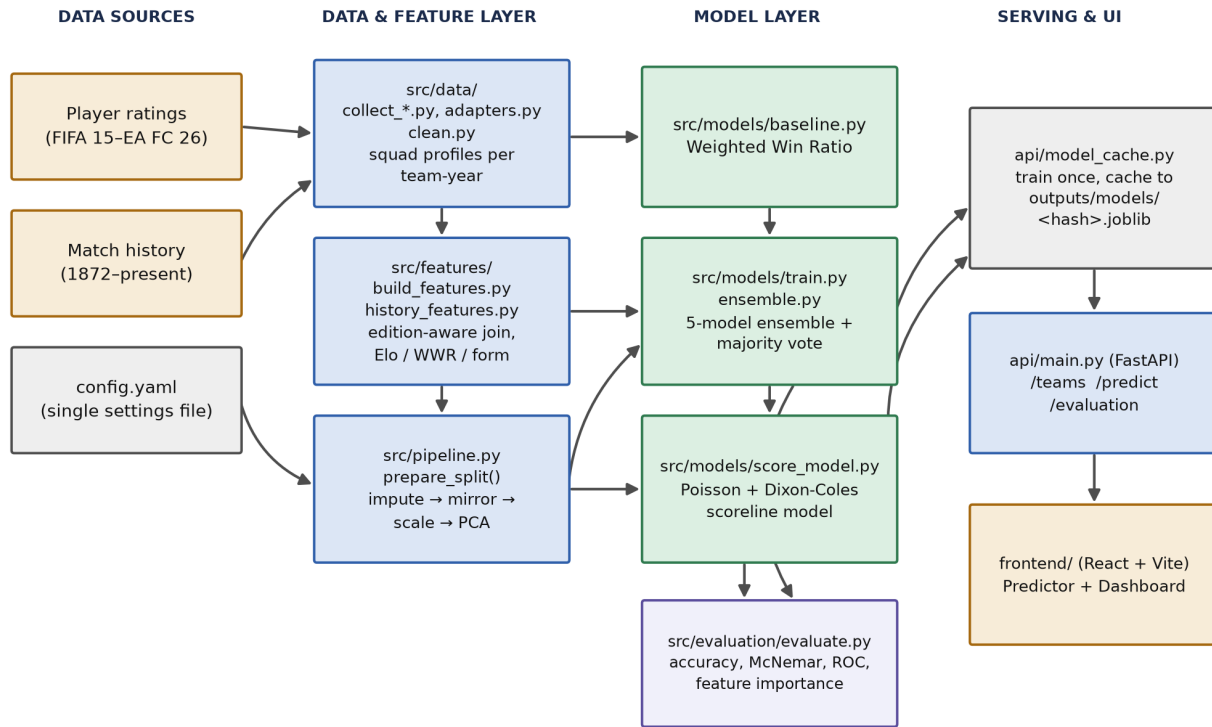


Figure 1. End-to-end system architecture, from raw data sources to the served web application. Each box names the module that implements it.

3.1 Design Principles

- Single configuration surface. Every tunable choice, data years, feature representation, model hyperparameter search space, holdout years, lives in one file, config/config.yaml, so a change to the modelling approach does not require touching code.
- Leakage resistance by construction. Historical features are computed strictly as of each match date rather than relying on the train/test split alone to prevent lookahead (Section 5.2); the held-out evaluation years are dropped before feature selection or tuning ever sees them (Section 7).
- Train once, cache, serve. Training the full pipeline takes several minutes; the API trains once per distinct configuration (keyed by a hash of the training-relevant config sections) and serves from a cached bundle on every subsequent start.
- Two tasks, one feature matrix. The classifier and the scoreline model are built from the same engineered features and the same train/test split machinery, differing only in the prediction target and in whether draws are dropped.

3.2 Component Overview

Table 1. Module responsibility by architectural layer.

Layer	Component	Responsibility
Data	src/data/collect_players.py collect_matches.py adapters.py	Downloads the raw player-rating and match-history datasets from Kaggle, and normalises the post-FIFA-23 EA Sports FC sources onto the same column schema as the earlier FIFA-branded files.

	src/data/clean.py	Selects each nation's squad per edition (real call-ups where tagged, a rating-based proxy otherwise) and aggregates player attributes into one profile per team per year.
Features	src/features/build_features.py	Joins team-year profiles onto each match using the FIFA edition current at the match date, producing paired and difference features and a neutral-venue flag.
	src/features/history_features.py	Maintains a running Elo, weighted-win-ratio, form, and head-to-head state per team, advanced match by match, and reads it back as of any target date.
Orchestration	src/pipeline.py src/score_pipeline.py	Builds the full match dataset once per run, then for each seed: splits, imputes, mirrors, scales, applies PCA, and hands the result to the model layer. The scoreline counterpart shares this machinery.
Models	src/models/baseline.py ensemble.py / train.py score_model.py	The Weighted Win Ratio baseline; the five-classifier ensemble with cross-validated hyperparameter search; the Poisson plus Dixon-Coles scoreline model.
Evaluation	src/evaluation/evaluate.py	Accuracy by scoring bucket, McNemar's test, ROC-AUC, vote-agreement breakdown, and PCA-mapped feature importance.
Serving	api/model_cache.py api/main.py	Trains and caches the full bundle keyed to configuration; exposes /teams, /predict, and /evaluation over FastAPI.
Presentation	frontend/ (React + Vite)	A live team-versus-team predictor and an evaluation-results dashboard, talking to the API through a dev-server proxy.

4. Data

4.1 Data Sources

Four datasets feed the project: three player-rating sources and one match-history source. EA renamed the FIFA series to EA Sports FC after the FIFA 23 edition, and the original player-dataset publisher stopped releasing new editions at that point, so the 2025 and 2026 editions had to be sourced from different publishers and mapped onto the same schema (Section 4.2).

Table 2. Data sources by year range.

Years	Dataset	Source	File
2015–2024	FIFA Complete Player Dataset	Kaggle: stefanoleone992 [5]	players_<edition>.csv
2025	EA Sports FC 25 database	Kaggle: nyagami [6]	male_players.csv
2026 †	EA Sports FC 26 player ratings	Kaggle: justdhia [7]	ea_fc26_players.csv
1872–present	International football results	Kaggle: martj42 [8]	results.csv

† Wired into the collection script but not exercised in this project: *data.years* stops at 2025 in the current configuration, and the 2026 source's column schema has not been checked against a real download.

4.2 The FIFA-to-FC Transition

The switch in data source at the 2025 edition is not a cosmetic detail: the held-out season that this report's headline result is measured on (Section 8.1) is built entirely from the FC 25 source, and it differs from the earlier years in two ways that matter for interpreting that result.

- No national-team call-up tag. The FIFA-era files carry a `nation_position` field, populated only for players actually named to a squad that edition, which lets the cleaning step build an accurate roster. Neither FC-era source carries an equivalent field, so every 2025 (and, if used, 2026) squad falls back to the rating-based proxy described in Section 4.3, rather than a confirmed call-up list.
- Goalkeeping attributes are not built the same way. The FIFA-era files rate every outfield player on goalkeeping attributes too, at a low value; the FC-era sources populate those attributes only for actual goalkeepers. A 2025 profile's goalkeeping mean therefore reflects only its keepers, where a 2019 profile's reflects the whole squad. This is arguably the more correct behaviour, but it is a real distribution shift landing exactly at the boundary between the development data and the held-out season.

Neither issue invalidates the headline result, but both are reasons to read “82.2% on unseen 2025 matches” as “on proxy squads with a goalkeeping-feature discontinuity” rather than as a clean like-for-like extension of the earlier years.

4.3 Squad Selection and Team-Year Profiles

Each nation's squad for a given edition is built in one of two ways. Where the source carries a national-team call-up tag, the real roster is used directly. Where it does not, either because the federation was one of the roughly one third EA does not license in a given FIFA-era edition, or because the whole edition is FC-era, a proxy squad is built instead: the highest-rated available players, filled against a positional quota rather than taken as an unconstrained top-23. The quota exists because taking the top 23 by overall rating outright was tried first and found to be quietly broken: EA rates goalkeepers systematically below outfield players, so a nation's best goalkeeper is often well outside its own top 23 by that metric alone (Algeria's best-rated 2017 goalkeeper ranked 32nd in the squad by overall rating), and 58% of the resulting proxy squads contained no goalkeeper at all. That does not only corrupt the goalkeeping features; it skews every squad-mean attribute toward outfield positions, silently, for well over half the affected teams.

Team names are also reconciled between the player dataset and the match-history dataset before the two are joined; the two sources disagree on the spelling or form of several nations (for example, the player dataset's “Korea Republic” against the match dataset's “South Korea”), and a small alias table in the cleaning module normalises these before any join happens.

Each team-year profile is built from squad means across the technical, physical, mental, and goalkeeping attribute groups, together with shape statistics, the top three players' average rating and the spread of overall ratings within the squad, and per-position aggregates. The shape statistics were added after a specific failure case: an early version of the model gave Brazil a 78% win probability over a Norway side that actually won 2-1, driven almost entirely by the squad mean (Brazil 84.4 against Norway's 76.6), while the gap by top-three average was only 1.7 points and Norway's best player (Haaland, rated 91) was rated above Brazil's. The squad mean measures depth, which matters for a league season; a single knockout match is often decided by the best two or three players on the pitch, which the mean cannot see on its own.

4.4 Data Scale

Table 3. Dataset scale after cleaning and joining.

Quantity	Value
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Team-year profiles built	1,749, across 189 national teams
Usable matches (finals + qualifiers, 2015–2025)	1,504
Development matches (2015–2024, after holdout withheld)	1,223
Held-out 2025 matches	281 (qualifiers only; no 2025 World Cup finals)
Raw engineered features (paired + difference representation)	up to 188, depending on year scope
Feature columns used by the classifier (difference-only representation)	64
Principal components retained (95% variance threshold)	approximately 22–23

5. Feature Engineering

5.1 Match-Level Profile Features

Because every match compares two teams, each profile attribute is represented three times in a raw feature row: once for Team A, once for Team B, and once as their difference. The difference representation is what the paper's non-redundancy principle argues for, and it was confirmed rather than assumed: the paired and difference columns are collinear by construction, and dropping to the difference alone (Section 5.5) held accuracy while noticeably reducing seed-to-seed variance.

The join itself is edition-aware rather than calendar-year-aware: a FIFA edition labelled year Y ships around September of year Y-1 and describes that season's squads, so it is the edition actually current from September of Y-1 through August of Y. Joining on a match's raw calendar year instead means an October fixture reads ratings published thirteen months earlier while a newer edition already exists, and roughly two thirds of the matches in this project's scope are played between September and December. Correcting the join to use the edition current at the match date moved 55% of match rows onto a newer set of ratings and was, on its own, the single largest improvement measured over the course of the project (Section 8.3).

A neutral-venue flag is attached to every row from the match-history data. World Cup finals are played almost entirely at neutral venues, but the training data is overwhelmingly non-neutral qualifiers, where the designated home side wins substantially more often than a team on neutral ground does. Without the flag, the model has no way to distinguish the two situations and applies a phantom home advantage to every neutral-venue prediction, which is every prediction the system actually needs to make at the tournament itself.

5.2 Historical Match Features

A second, independent feature family captures each team's historical form, built from a running state advanced match by match over the full international record (not restricted to World Cup matches, since Elo and recent-form estimates need volume to be meaningful). Every value is computed strictly from matches earlier than the one being scored, including an explicit same-day exclusion so that two fixtures played on the same date cannot see each other's results regardless of the order they appear in the underlying file. This makes leak-freedom a structural property of the feature construction rather than a property that depends on which split strategy is used downstream.

Table 4. Historical match-feature families, each emitted as an `_a / _b / _diff` triple per team.

Feature family	Description
----------------	-------------

elo_*	Classic Elo rating, updated chronologically. K is flat (not competition-weighted, see Section 6.2's note on this) with a goal-difference multiplier and a home-advantage adjustment applied only on non-neutral ground.
wwr_*	The baseline's own credibility-weighted win ratio, exposed as a classifier feature too, so the ensemble has access to the same signal the baseline is built from.
played_*	Matches played to date, the credibility term the WWR baseline shrinks its estimate on.
h2h_winrate_*	Head-to-head record against this specific opponent.
form_win_* / form_gd_*	Win rate and mean goal difference over each team's most recent matches (a configurable window).

5.3 Neutral-Venue Row Mirroring

Because which team is recorded as “home” (Team A) is essentially arbitrary for a neutral-venue match, the training set is augmented by appending a side-swapped copy of every neutral-venue row with the label flipped, so the model cannot learn to prefer whichever side happens to be listed first. Mirroring was tried unrestricted first and found to cost accuracy, since asserting that the away side of an ordinary home-and-away qualifier had home advantage is simply false; restricted to the rows where the ordering genuinely is arbitrary, it is a net positive and meaningfully reduces seed-to-seed variance.

5.4 Squad-Size Reliability

Team-year profiles are built from whoever the rating source actually rated for that nation and year, which for smaller federations can be a handful of players rather than a full squad. Roughly a third of match rows have at least one side built from fewer than five rated players, and close to half have a side under eleven. A minimum squad-size feature is carried so the model can see how much squad actually backs a given comparison; filtering out thin-squad rows entirely was tried and found to hurt accuracy at every threshold tested, since the training volume lost costs more than the noise removed. The feature is kept because it is a single column that genuinely describes the reliability of the row it sits in, even though it did not move accuracy on its own.

5.5 Scaling and Dimensionality Reduction

Before the classifiers see the data, features are imputed from the training split's own mean, standardised, and passed through Principal Component Analysis, retaining enough components to explain 95% of variance. All three steps are fit on the training rows of the specific split being used and applied unchanged to the corresponding test rows, so no information from a test row ever influences how a training row is scaled or imputed. Figure 2 shows the resulting variance curve for one representative split.

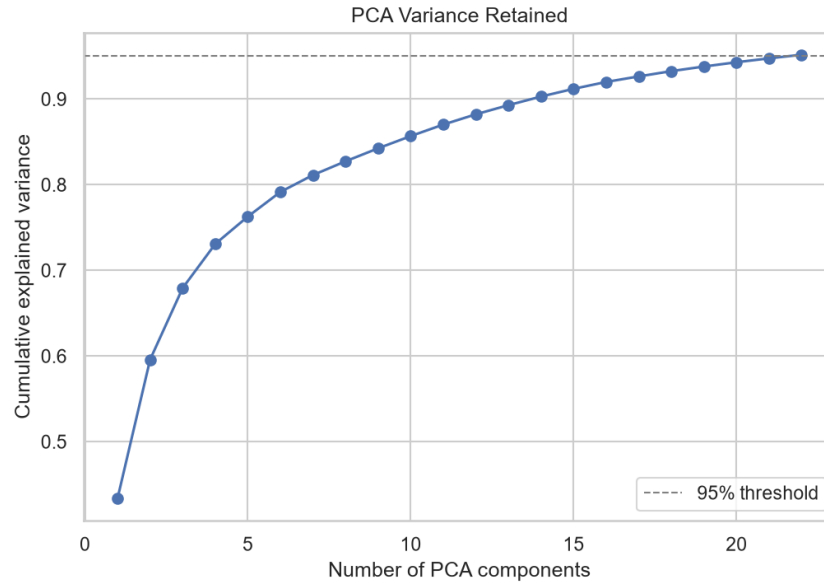


Figure 2. Cumulative explained variance against number of principal components, for one representative seed. The 95% threshold is typically reached at approximately 22–23 components.

5.6 Final Feature Set

After settling on the difference-only representation, the classifier's raw feature matrix has 64 columns, drawn from the squad-profile differences described in Section 5.1 and the six historical feature families described in Section 5.2, plus the neutral-venue flag and the minimum squad-size feature. This is compressed by PCA to roughly 22–23 components before reaching the model layer.

6. Modeling

Figure 3 shows one full seed of the training cycle: the same engineered dataset feeds both the baseline and the ensemble, and separately feeds the scoreline model, with the ensemble and baseline compared on identical test rows.

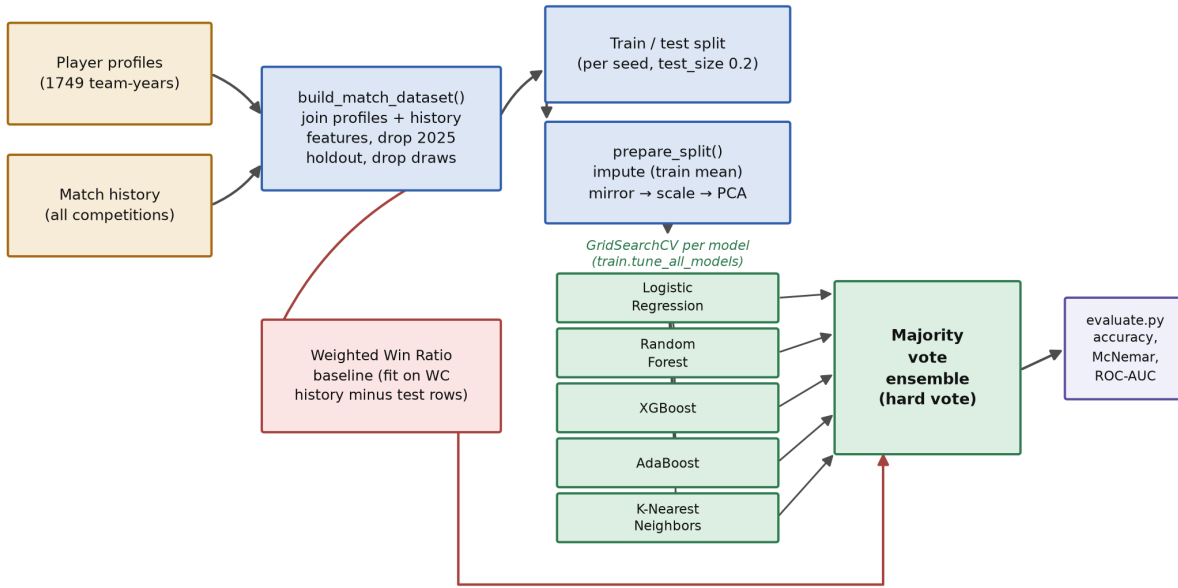


Figure 3. One seed of the training and evaluation cycle. The whole cycle is repeated for five fixed seeds; only the split and the model fits vary between them, since the engineered dataset itself is built once and reused.

6.1 Baseline: Weighted Win Ratio

The baseline reproduces the reference paper's [1] Weighted Win Ratio model. Where two teams have met at least a configured minimum number of times, their head-to-head record decides the prediction directly. Below that threshold, each team's win record is shrunk toward the global average by a credibility-weighted estimator, the same family used for IMDB's weighted ratings:

$$WWR = \frac{v}{v + m} R + \frac{m}{v + m} C$$

where v is the number of matches the team has played, R its own win ratio, C the global win ratio across all teams, and m a shrinkage constant (set to 10). A team with very few recorded matches is pulled toward the global average; a team with a long track record is trusted close to its own ratio. The baseline is fit on World Cup history with the specific test rows for that split removed, which mirrors how the classifier is evaluated but is worth stating explicitly, since a reader could otherwise assume the baseline only ever sees historical, pre-tournament data.

6.2 Ensemble Classifier

The proposed model is a majority-vote ensemble of five classifiers, each chosen for a distinct bias and variance profile: Logistic Regression as a linear, interpretable member; Random Forest and AdaBoost as tree-based methods that tend to perform well on tabular data; K-Nearest Neighbours to add a member with a genuinely different decision boundary; and XGBoost [4], a gradient-boosted tree implementation chosen for its typically strong performance on structured, tabular problems like this one. The first four are implemented via scikit-learn [3]. Each model is tuned independently by five-fold cross-validated grid search over the ranges in Table 5, and the five tuned models are then combined by a hard majority vote.

Table 5. Hyperparameter grids searched per model.

Model	Hyperparameters searched
Logistic Regression	$C \in \{0.01, 0.1, 1.0, 10.0\}$
Random Forest	$n_estimators \in \{100, 200, 400\}$; $max_depth \in \{None, 5, 10, 20\}$; $min_samples_leaf \in \{1, 2, 5\}$
XGBoost	$n_estimators \in \{100, 200, 400\}$; $max_depth \in \{3, 5, 7\}$; $learning_rate \in \{0.01, 0.05, 0.1\}$
AdaBoost	$n_estimators \in \{50, 100, 200\}$; $learning_rate \in \{0.5, 1.0, 1.5\}$
K-Nearest Neighbors	$n_neighbors \in \{3, 5, 7, 11, 15\}$; $weights \in \{uniform, distance\}$

One methodological note affecting the Elo feature deserves mention here since it was measured as part of tuning the overall system: a competition-weighted Elo update (higher K for World Cup matches than for friendlies), the convention used by the widely cited World Football Elo Ratings [9], was tried and found to perform slightly worse than a flat update rate across four of five seeds. Tying the update rate to competition importance makes the rating move faster on exactly the matches that already have separate recent-form features covering them; a flat rate gives the ensemble the stable, long-run reputation signal that Elo is actually useful for. A goal-difference multiplier on the Elo update, by contrast, was kept, since margin of victory is real evidence no other feature in the set carries.

6.3 Model Selection Protocol

For each of five fixed random seeds, the engineered dataset is split into train and test partitions (80/20), the training partition is imputed, mirrored, scaled, and reduced by PCA as described in Section 5, each of the five classifiers is tuned by cross-validated grid search on that same training partition, and the tuned ensemble is then scored once on the held-out test partition for that seed. Repeating this five times, rather than relying on a single split, is what makes it possible to report a variance alongside the accuracy figure and to say with more confidence that an improvement is not simply an artifact of one lucky split.

6.4 Scoreline Model

A secondary model predicts a full scoreline rather than a single win/lose label, using exactly the same feature matrix as the classifier so that it generalises the existing task rather than sitting beside it as unrelated work. Two Poisson regressions estimate each side's expected goals, and a Dixon-Coles correction [2] is applied to the resulting grid to fix a specific, measured shortcoming of independent Poisson scoring: it under-predicts draws (21.8% against an actual rate of 24.2%) because it cannot represent any dependence between the two sides' scores at low scorelines. Two modelling choices behind this were checked rather than assumed. A negative binomial distribution, the standard response to the raw goal counts looking overdispersed, turned out to be unnecessary: once the engineered features are conditioned on, the dispersion is almost exactly 1, so the apparent overdispersion in the raw counts is explained by the features already in the model. And the Dixon-Coles correction is a calibration fix rather than an accuracy gain: it improves draw calibration and is marginally better on log-loss and the ranked probability score, but it cannot change win/lose accuracy at all, since it is symmetric between the two sides, and it costs a small amount of exact-scoreline accuracy by making low-scoring draws the modal prediction more often.

Because it shares the classifier's feature matrix, the scoreline model's derived win/lose accuracy can be compared directly against the classifier's own: on development data it reaches $79.18\% \pm 2.41\%$ against an untuned ensemble's $78.29\% \pm 1.62\%$ on the same splits, evidence that predicting goals costs nothing on the original binary task while additionally representing the roughly 24% of matches that end in a draw, which the binary framing cannot express at all.

7. Evaluation Methodology

7.1 Why a Formal Holdout Was Introduced

Over the course of this project, on the order of fifteen distinct feature and model configurations were compared against the same five random splits, each kept or discarded according to how it scored there. That is model selection performed directly against evaluation data, and its effect compounds quietly rather than announcing itself: across that same stretch of work, the reported random-split accuracy climbed from 78.9% to 80.3%, while an untuned temporal split (train on everything before a fixed date, test on what came after) moved in the opposite direction, from 79.7% down to 79.5%. The gains being reported on the random split had simply stopped replicating on a split that was not being selected against.

In response, the most recent full year of data, 2025, was withheld entirely from development. It never reaches feature selection, hyperparameter tuning, the five-seed variance figures, or the cached model bundle the API serves; the only code path that reads it is a dedicated script that trains on all development data and scores the held-out season exactly once. The cost of doing this is real and is reported alongside the benefit: 281 fewer training matches, which measurably lowers accuracy on other held-out checks (Section 8.10). The benefit is that the headline figure in Section 8.1 is a number no development decision was made against.

One caveat is kept alongside that benefit rather than hidden behind it: the holdout was locked after most configuration choices had already been made with 2025 visible, so it is clean for every change made from the point the FC 25 data adapter was added onward, but not for the full history of the project's development. That the held-out score came in above the development estimate, rather than below it, is the reassuring direction for that caveat to point in, though it remains a single measurement of a single season.

7.2 Development, Holdout, and Backtest Periods

Figure 4 lays out the three periods of data referred to throughout this report and what each one is used for.

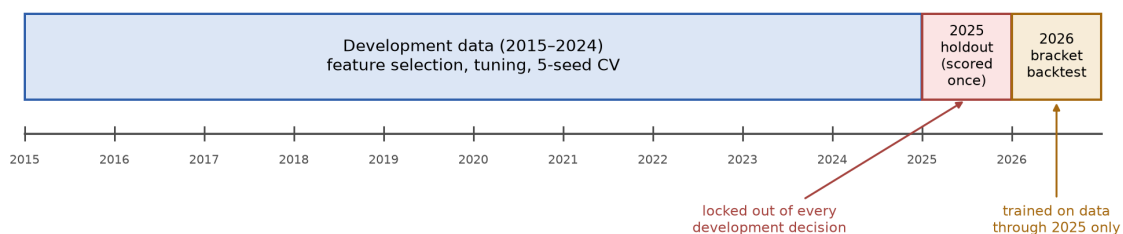


Figure 4. What each period of data is used for. The 2025 holdout never reaches tuning or the seed-variance figures; the 2026 backtest is scored using only information available before that tournament's kickoff.

A third, independent check uses the real 2026 World Cup knockout bracket. Team strength for that backtest comes from the EA FC 25 squad ratings, published before the tournament began, and the historical features are cut at that tournament's kickoff date, so no 2026 result can inform a 2026 prediction. This makes the bracket a genuine out-of-sample test rather than a cross-validation fold, and it is trained on data that excludes the 2025 season as well, so it is not merely a repeat of the holdout check under a different name.

7.3 Metrics

- Overall accuracy, and accuracy split by high-scoring versus low-scoring matches, to check whether either model's edge is concentrated in a particular kind of game.

- Precision, recall, and F1, since accuracy alone can hide the class imbalance between home and away wins.
- ROC-AUC, for the ensemble's ranking quality independent of the 50% decision threshold.
- McNemar's test on paired predictions, to assess whether one model's advantage over the other on the same matches is larger than sampling noise would produce.
- Vote-agreement breakdown (how often the ensemble is right when 3, 4, or 5 of its five members agree), and PCA-mapped feature importance from the two tree-based members, approximate because principal components have no direct real-world meaning of their own.

8. Results

8.1 Headline: The Withheld 2025 Season

Trained on all 1,223 development matches (2015–2024) and scored once on the 281 withheld matches of 2025:

Table 6. Headline accuracy on the withheld 2025 season.

Model	Accuracy
Proposed ensemble	82.2%
Weighted Win Ratio baseline	75.4%
Always predict the home side	59.4%

The 6.8 percentage-point margin over the baseline is statistically significant: McNemar's test on the paired predictions gives $p = 0.0134$ (36 matches the ensemble won and the baseline lost, against 17 the other way), the first time that has been true at any point in this project. Individual base models before the vote score between 79.4% and 82.6%, so the ensemble sits at the top of its own members' range rather than clearly beyond it. Table 7 gives the supporting metrics on the same 281 matches, covering each of the measures set out in Section 7.3.

Table 7. Supporting metrics on the 281 held-out 2025 matches.

Metric	Ensemble	Baseline
Overall accuracy	82.2%	75.4%
Accuracy, high-scoring / low-scoring	86.1% / 77.2%	78.5% / 71.5%
Precision / recall / F1	0.842 / 0.862 / 0.852	0.827 / 0.743 / 0.782
ROC-AUC	0.907	not applicable
Challenging cases recovered	36 of 69 (52.2%)	not applicable
Unanimous vote (5 of 5), 242 matches	85.1%	not applicable
Split vote (3 or 4 of 5), 39 matches	64.1%	not applicable

The baseline produces a hard prediction rather than a ranked probability, so ROC-AUC and the vote-agreement rows do not apply to it.

Two rows are worth drawing out. Recall (0.862 against the baseline's 0.743) is where most of the ensemble's advantage sits: the baseline misses substantially more genuine home wins, which is consistent with it falling

back on a shrunken win ratio whenever a pairing has little head-to-head history. And the gap between a unanimous vote (85.1% correct) and a split one (64.1%) means the ensemble's own internal disagreement is itself a usable confidence signal, a point Section 8.4 returns to on development data.

Two qualifications belong next to this number. All 281 holdout matches are World Cup qualifiers; no finals were played in 2025, so the headline is measured on qualifying fixtures only, as, largely, is the whole project (finals are 100 of the 1,504 usable matches). And the 2025 squads behind it are entirely rating-based proxies with a goalkeeping-feature discontinuity relative to earlier years (Section 4.2).

8.2 Development Results Across Five Seeds

On development data alone, averaged over five fixed random seeds, the ensemble scores $79.0\% \pm 3.3\%$ against the baseline's 76.0%, ahead of the baseline on every one of the five seeds. Per-seed McNemar significance is not reached on any individual seed (p between 0.188 and 0.617, with roughly 245 test rows per seed); a three-point gap on that many rows is simply underpowered for the test. The supporting evidence at this stage of the project is not any single seed's p-value but that the margin is positive across all five, and it is the larger, better-powered holdout margin in Section 8.1 that actually reaches significance, as Figure 5 shows for each individual seed.

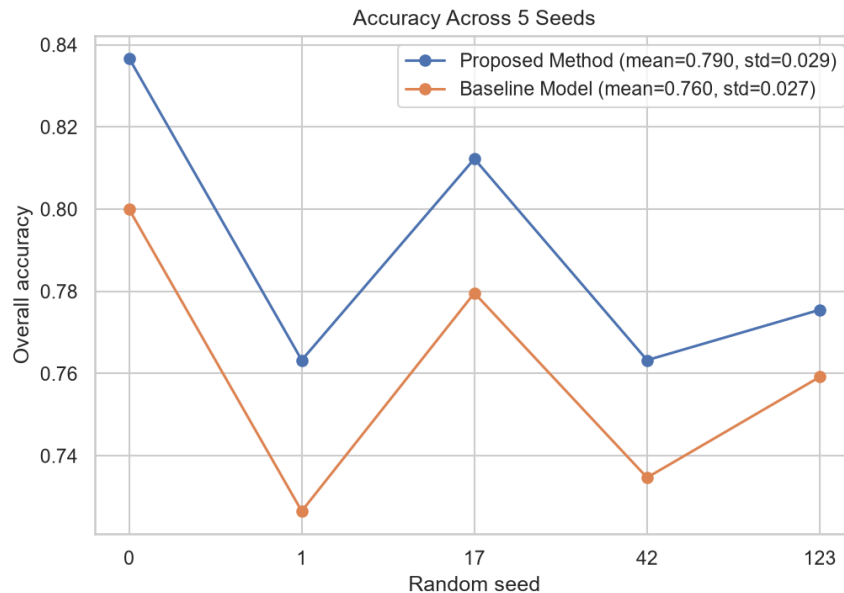


Figure 5. Ensemble and baseline accuracy across the five development seeds. The ensemble leads on every seed; the spread across seeds is the variance quoted alongside the mean.

8.3 How the Model Reached This Result

The ensemble did not start ahead of the baseline. Its first working version scored $75.5\% \pm 2.5\%$ against the baseline's 74.8%, ahead on only two of five seeds, because the two models were built from disjoint information: the ensemble knew squads and nothing about results, the baseline knew results and nothing about squads, and the two disagreed on roughly a third of matches. Each change below was measured independently across all five seeds before being kept.

Table 8. Development accuracy after each change, measured independently and kept in sequence.

Change	Accuracy	Margin over baseline	Seeds ahead
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Starting point	75.5% $\pm$ 2.5%	+0.7 pp	2 / 5
+ neutral_site feature	76.3% $\pm$ 2.3%	+1.5 pp	3 / 5
+ as-of-date history features	78.1% $\pm$ 3.0%	+3.3 pp	4 / 5
+ neutral-only row mirroring	78.5% $\pm$ 2.0%	+3.7 pp	5 / 5
+ squad shape & position aggregation	78.6% $\pm$ 1.8%	+3.8 pp	5 / 5
+ goal-difference-weighted Elo	78.9% $\pm$ 1.9%	+4.2 pp	5 / 5
+ difference-only representation	78.8% $\pm$ 1.1%	+4.0 pp	5 / 5
+ edition-aware squad join	80.3% $\pm$ 3.2%	+5.3 pp	5 / 5
+ EA aggregate ratings (FC 25 adapter) †	79.0% $\pm$ 3.3%	+3.0 pp	5 / 5

† Measured on development data only, after the 2025 holdout was withheld, so this row is not directly comparable to the rows above it, which were measured on the full set before the holdout existed. On the same development-only data, the previous configuration scores 78.6%, so the adapter itself is worth about +0.4 pp.

The single largest gain, the edition-aware squad join (Section 5.1), came from recognising that a FIFA edition ships roughly a year before the calendar year it is named for, so joining on calendar year alone was quietly serving most matches ratings that were, on average, a year stale. Two changes that looked reasonable in isolation did not survive measurement: filtering out thin-squad training rows cost accuracy at every threshold tried, and competition-weighted Elo underperformed a flat update rate on four of five seeds (Section 6.2).

8.4 Per-Model and Vote-Agreement Breakdown

Figure 6 breaks accuracy down by individual base classifier, before the majority vote is applied.

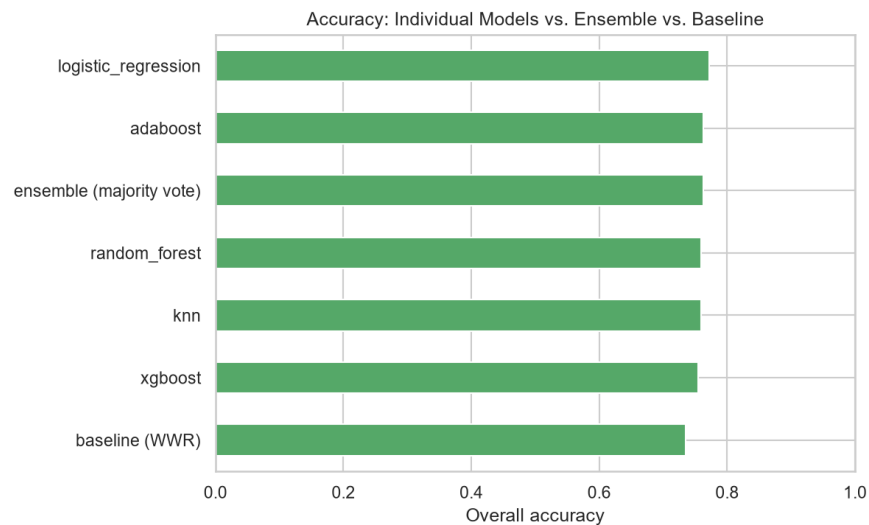


Figure 6. Accuracy of each of the five base classifiers before the majority vote is applied, one representative seed.

Pooled across the five seeds, matches where all five base classifiers agree are correct 83.9% of the time; matches decided by a narrower majority are correct 56.2% of the time, close to a coin flip. This is a useful signal in its own right: the ensemble's own internal disagreement is informative about how much to trust a given prediction, independent of the predicted probability itself.

8.5 Confusion Matrices

Figure 7 gives the full breakdown of correct and incorrect predictions by class for both models on the same seed as the metrics above.

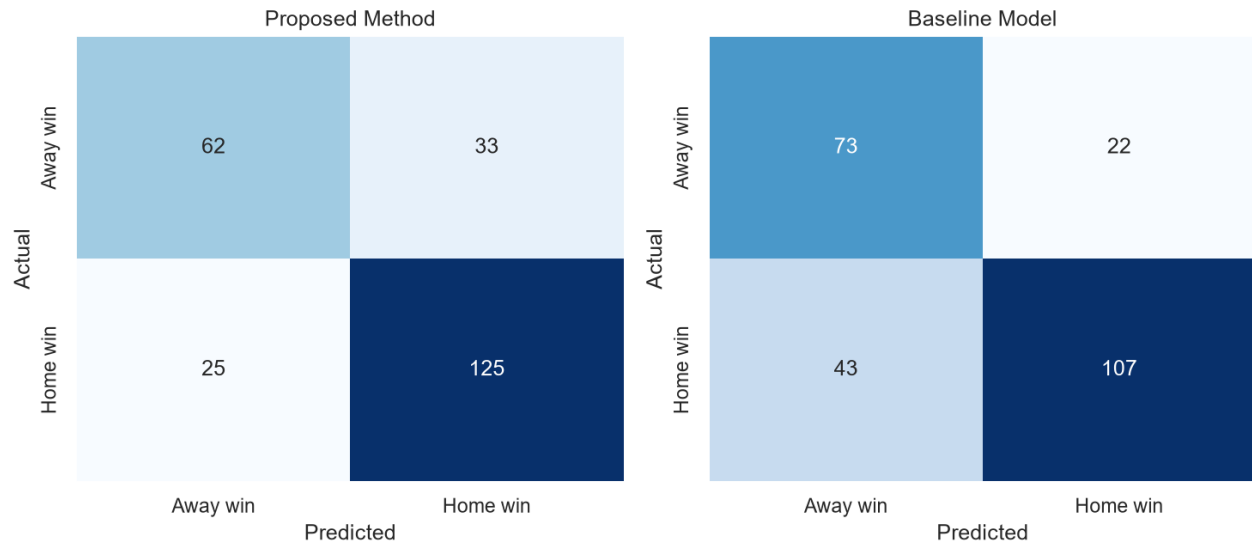


Figure 7. Confusion matrices for the proposed ensemble and the baseline, one representative seed, over the classes {away win, home win}.

8.6 ROC Curve

Figure 8 gives the ensemble's ROC curve, independent of the 50% decision threshold used elsewhere in this report.

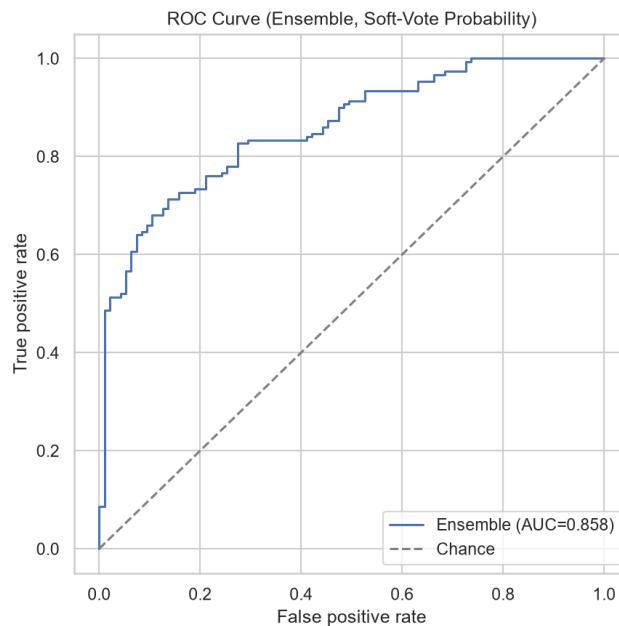


Figure 8. Ensemble ROC curve, one representative seed.

8.7 Feature Importance

Feature importance is taken from the two tree-based ensemble members (Random Forest and XGBoost) and mapped back from PCA-component space to the original 64 engineered features. This mapping is an approximation rather than an exact attribution, since principal components are linear combinations of the original features and have no direct real-world meaning of their own; it is reported as a directional signal about which feature families matter, not as a precise ranking. Figure 9 gives the top fifteen by this measure.

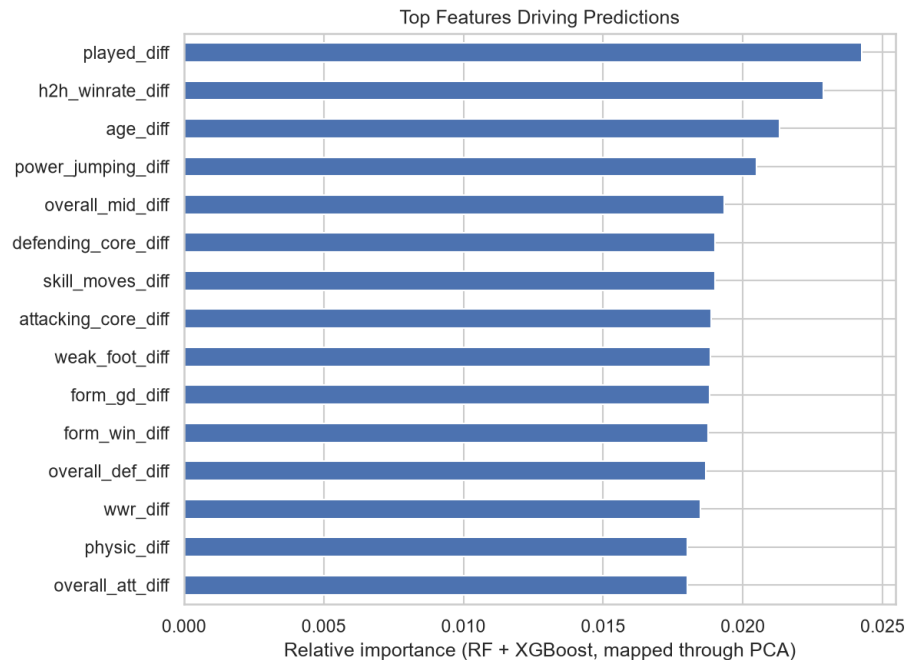


Figure 9. Top engineered features by importance, PCA-mapped, one representative seed.

8.8 Scoreline Model Results

The scoreline model of Section 6.4 is evaluated on the same development splits as the classifier, against two reference points: an untuned version of the classifier ensemble on identical splits, and a league-average floor that predicts the same goal distribution for every fixture regardless of the teams involved. Figures are averaged over the same five seeds used throughout Section 8.2.

Table 9. Scoreline model results against reference floors, five-seed development average.

Metric	Scoreline model	Reference point
Derived win accuracy	79.18% $\pm$ 2.41%	78.29% $\pm$ 1.62% (untuned ensemble)
Ranked probability score (RPS)	0.1683	0.2283 (league-average floor)
Exact scoreline accuracy	15.3%	10.4% (modal-scoreline floor)
Goal mean absolute error	0.87	1.01 (floor)

The first row carries the most weight. Derived win accuracy is what the scoreline model scores on the original binary task once its grid is collapsed to P(home win), and at 79.18% it matches the classifier ensemble on the same splits rather than trailing it. Predicting a full scoreline therefore costs nothing on the task the project set out to solve, while additionally representing the roughly 24% of matches that end in a draw, which the binary framing cannot express at all. RPS and goal error both sit well clear of the no-skill floor, so the model is

genuinely conditioning on the features rather than reproducing a league-average scoreline.

The Dixon-Coles correction behaves exactly as Section 6.4 predicts. On the served seed it moves the predicted draw rate from 21.7% to 22.9% against an actual 24.1%, closing about half the gap, and it leaves derived win accuracy completely unchanged (77.55% either way on that seed), confirming that a side-symmetric correction cannot move a home-versus-away comparison. It is kept for calibration, and it is deliberately not claimed as an accuracy gain.

8.9 Validity Checks

Two conditions that would have invalidated the headline figure were tested directly rather than assumed to hold.

- Is the random split hiding that the model trains on the future? Splitting temporally instead, training on everything up to a fixed date and testing only on matches after it, gives $79.7\% \pm 0.7\%$ against the baseline's 77.0%. Accuracy does not drop under a temporal split, so the random-split figures reported elsewhere in this report are not inflated by hindsight.
- Is the baseline handicapped relative to the ensemble? The ensemble's historical features are built from the full international record (roughly 49,500 matches) while the Weighted Win Ratio baseline is fit on World Cup matches only (roughly 9,800), a five-times information advantage for the ensemble. Giving the baseline the full record as well moves it from 74.8% to 74.9%, a real but small effect: the like-for-like margin at that measurement point is +4.0 percentage points rather than +4.2, which does not change the project's conclusions.

8.10 Held-Out Test: The 2026 World Cup Knockout Bracket

Table 10. Accuracy by knockout round, 2026 World Cup bracket.

Round	Ties	Ensemble	Baseline
Round of 32	16	88%	50%
Round of 16	8	88%	88%
Quarter-finals	4	100%	100%
Semi-finals	2	0%	50%
Overall	30	83.3%	66.7%

At 83.3%, this out-of-sample figure sits above the development estimate of $79.0\% \pm 3.3\%$, the main evidence that the model generalises rather than having been fit to its own cross-validation splits, though “above” is worth treating cautiously rather than celebrating. The goal-difference-weighted Elo change (Section 6.2) was worth an estimated +0.33 percentage points in cross-validation and appears worth roughly ten times that here; that gap is three matches out of thirty, so most of it is small-sample variation, and the cross-validated figure remains the one to trust.

With only 30 ties, the 95% interval is approximately ± 13.3 percentage points, so the 16.6-point margin over the baseline is not itself statistically significant. Four ties were decided on penalties, which the model has no mechanism to represent; it predicts a ninety-minute winner and is scored against the shootout result. It misses both semi-finals (favouring France over Spain and England over Argentina), and it is trained without any 2025 matches, since they are held out, costing it an estimated one tie's worth of accuracy.

Two fixtures in this bracket are not backtests at all. Neither the third-place playoff nor the final has a recorded score in the dataset, so the model's predictions for them are genuine forward predictions, made on information

available before the tournament and not yet scored against anything:

- Third-place playoff: **England** to finish third, at 62% confidence.
- Final: **Spain** to beat Argentina, at 52% confidence. That is barely above a coin flip, and it is the appropriate amount of conviction for a final between those two sides; a model returning 80% there would be describing its own overconfidence rather than the match.

9. System Implementation

9.1 Backend API

A FastAPI application exposes three endpoints. GET /teams lists every (team, edition) pair that has a squad profile, which the frontend uses to populate its team pickers and to restrict the edition selector to years both chosen teams actually share. POST /predict scores a single matchup: it returns the ensemble's winner and per-model votes, and, where the scoreline model is enabled, an expected-goals and outcome-probability breakdown from it as well. Because World Cup matches are played at neutral venues but the model is trained on rows carrying a nominal home side, a request can ask to have the fixture scored in both team orderings and averaged, which removes the model's otherwise real sensitivity to which team happens to be passed first. GET /evaluation returns everything the results dashboard renders, computed from the cached model bundle with no request parameters.

All three endpoints depend on a module-level state populated once, during the application's startup handler, by loading or training the model bundle. A cache miss takes several minutes, during which the server does not accept traffic in any meaningful sense; the cheapest endpoint that only succeeds once startup has finished is what the startup script and the frontend both poll on.

9.2 Model Caching Strategy

Training the full pipeline, five classifiers across five seeds plus the scoreline model, is expensive enough that it should not happen on every server restart. The caching layer hashes the training-relevant sections of the configuration file together with the fixed seed list into a short key, and reuses a previously trained bundle from disk whenever that key has not changed; changing an unrelated setting, such as the project's display name, does not invalidate the cache, while changing any modelling knob does.

9.3 Frontend Application

Figure 10 traces a single request from the browser through to the trained model bundle.



Figure 10. Request path from the browser to the trained model bundle. The frontend never hardcodes the backend's origin, so the same build works against the API on any local port.

The web application is a React and Vite single-page app with two panels: a live team-versus-team predictor and an evaluation-results dashboard, both driven entirely by the three API endpoints described above. Because the backend can take several minutes to finish training on a cache miss, the frontend treats a server error without an accompanying error message as “still starting” rather than as a failure, and polls until the backend responds properly or returns a genuine, message-bearing error such as an unknown team name.

9.4 Running the System Locally

A single command starts both halves of the application, waits for the backend to finish training or loading from cache, and opens a browser window once the system is ready. The backend and frontend can also be run independently for development, and a companion script stops whatever the start script left running, since a second invocation that finds both expected ports already in use exits immediately with nothing left to interrupt.

10. Testing

As of this report, the suite comprises 121 tests across the seven files below, all passing (*pytest tests/ --ignore=tests/test_api.py*, verified 2026-09-19). It is concentrated on the components most likely to silently produce an incorrect result rather than an obvious crash: the pieces that decide labels, decide who is in a squad, and decide what a feature actually measures.

Table 11. Test file coverage.

Test file	Covers
test_baseline.py	The Weighted Win Ratio formula and its head-to-head tie-breaking.
test_build_features.py	The paired/difference feature join, draw dropping, label correctness, and error handling for an unknown team or year.
test_clean_profiles.py	filter_national_team_squad's licensed-versus-proxy branches and the positional-quota backfill.
test_history_features.py	Elo update arithmetic, weighted-win-ratio parity with the baseline, as-of-date snapshotting, and same-day exclusion.
test_score_model.py	Dixon-Coles mass preservation, the Poisson scoreline model's fit and predict paths, and RPS/log-loss sanity checks.
test_evaluate.py	McNemar's test and the PCA-to-original-feature importance mapping.
test_api.py	End-to-end: boots the FastAPI application and exercises /teams, /predict, and /evaluation.

Known gaps: there is no test asserting that training a given seed twice produces bit-for-bit identical output, and the end-to-end API test currently accepts a 400 error response as a passing result on /predict regardless of whether that specific pair of teams should have succeeded, which would let a real regression in that endpoint pass silently. Neither gap affects the validity of the results reported in Section 8, since both concern test coverage rather than the modelling pipeline itself, but both are worth closing before the test suite is relied on as a full safety net.

11. Limitations

11.1 Data and Scope

- Player quality is a video-game rating, not an independent measure of ability, and the model inherits whatever bias that rating process carries, including uneven scouting depth across federations.
- Every 2025 squad, and every squad from an unlicensed federation in earlier years, is a rating-based proxy rather than a confirmed roster (Sections 4.2–4.3).
- The headline result is measured on qualifying matches only; only 100 of the 1,504 usable development-plus-holdout matches are World Cup finals at all.
- The model has no mechanism to represent a penalty shootout and is scored against the ninety-minute result when a tie is actually decided on penalties.
- Historical features mix friendlies, qualifiers, and finals rather than being tournament-specific, a deliberate choice made because Elo and recent-form estimates need match volume to be meaningful (Section 5.2).

11.2 Methodology

- The Weighted Win Ratio baseline's tie-break rule favours whichever team is passed as Team A, which very slightly flatters it on ties since home wins are the majority class.
- PCA-mapped feature importance (Section 8.7) is a best-effort approximation, not an exact attribution, since principal components have no direct real-world meaning.
- Most configuration choices in this project were made before the formal holdout existed, with the 2025 season visible; the holdout is clean for changes made from the FC 25 data adapter onward, not for the project's full development history (Section 7.1).
- Per-seed statistical power is thin: McNemar's test does not reach significance on any individual development seed, and a single run's accuracy figure should be read as one sample from a real, measured spread rather than as the model's exact accuracy.

11.3 Engineering

- The evaluation seed list is duplicated across four locations in the codebase rather than defined once, which risks silent drift if one copy is updated and another is not.
- The active classifier list cannot safely drop Random Forest or XGBoost without a corresponding code change, since a downstream feature-importance step reads those two model names directly.
- There is no dedicated health or readiness endpoint; a request that lands before the model bundle has finished training raises an unhandled server error rather than a clear "still starting" response.
- CORS is currently configured permissively, which is appropriate for local development but would need tightening before any deployment reachable outside a single machine.
- There is no eviction policy for cached model bundles on disk; every distinct configuration tried leaves its own multi-megabyte file behind.

12. Future Work

- Extend the withheld-evaluation approach to additional seasons as new tournament cycles complete, building a rolling multi-year out-of-sample track record rather than the current single 2025 holdout and one-off 2026 bracket backtest.

- Add explicit modelling of penalty-shootout outcomes for knockout ties, or at minimum report shootout-decided ties separately from ordinary misses.
- Periodically re-measure competition-weighted Elo as more tournament data accumulates; it underperformed a flat update rate on the data available during this project, but that could change with volume.
- Explore a graph- or sequence-based representation of individual players as a successor to the current squad-level aggregate features, in the direction the literature survey identifies as underexplored for football specifically.
- Extend the Dixon-Coles scoreline model toward full bracket simulation (for example, Monte Carlo tournament simulation from single-match probabilities) rather than single-match expected scorelines alone.
- Consolidate the duplicated seed-list configuration into a single source of truth, add a proper health/readiness endpoint to the API, and tighten CORS ahead of any deployment beyond local use.

13. Conclusion

This project set out to test whether combining player-level squad composition with historical team form could out-perform a strong, results-only baseline on World Cup match prediction, and to do so in a way that would survive scrutiny of how the result was obtained. The final measurement, an 82.2% accuracy ensemble against a 75.4% baseline on 281 matches withheld from every development decision, with a statistically significant margin, supports that the combination works, and the project's own record of what was tried and rejected (Section 8.3) shows that the improvement did not come from any single change but from several independently-measured ones compounding: an edition-aware join that fixed a genuine data-alignment bug, a neutral-venue flag and restricted row mirroring that removed a home-advantage artifact, and squad-shape features that fixed a specific, diagnosable failure case.

Equally, the project's discipline about what it does not yet know, an underpowered per-seed significance test, a 2026 bracket result too small to be significant on its own, proxy squads standing in for real rosters throughout the held-out season, is treated in this report as part of the result rather than as a footnote to it. The system is delivered as a complete, working pipeline from raw data to a served web application, backed by a tested codebase, and the qualifications documented in Section 11 define the scope within which its headline number should be read.

14. References

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- [9] "World Football Elo Ratings," [eloratings.net](https://www.eloratings.net) (accessed 2026).

Datasets [5]–[8] were accessed through Kaggle for this project's data collection step (Section 4.1); citations record the dataset and host, not a claim of authorship over the underlying data. Software libraries not separately cited above (FastAPI, React, Vite, pandas, NumPy, SciPy) are used as standard tooling rather than as sources of the project's methodology.